

it. Thus, efficient communication is ensured, even when a part of the ISM band is saturated. This whole behaviour is known as adaptive frequency hopping (AFH). AFH significantly reduces the probability of collisions and allows Bluetooth

devices to function well even in very difficult radio environments.

As stated earlier, signal strength can greatly impact reliability and FEC is one of the techniques used to address this. But, in addition, be aware that radio receivers work best when the level of

the received signal stays within a certain range (known as the golden range). There is a Bluetooth feature called LE power control, which allows connected devices to request dynamic changes to transmission power levels so that the received signal strength stays within this golden range. This is particularly useful when the distances between devices vary during normal use.

Avoiding buffer overflow

To mitigate the reliability challenges of buffer overflow, the rate of flow of packets has to be managed. This can be done using a technique known as flow control, which is used in many different communication technologies.

The Bluetooth Low Energy L2CAP layer of the stack applies flow control when required and supports a variety of different flow control modes. For example, if you are using the enhanced attribute protocol at the top of the stack, then it uses the Enhanced Credit Based Flow Control Mode within the L2CAP layer, allowing the receiving device to communicate to the transmitting device its current capacity to receive packets. If that is zero, then the receiver's buffer is full and the transmitter will wait before resuming packet transmission again.

So, it is a dynamic flow control strategy, which prevents buffer overflow.

Bluetooth LE reliability for connectionless communication

In Bluetooth, connectionless communication is called advertising. In the basic case, copies of each packet are broadcast on each channel in random channel order. Their time intervals are solely controlled by the broadcasting device, without any coordination with the transmitter. There is also

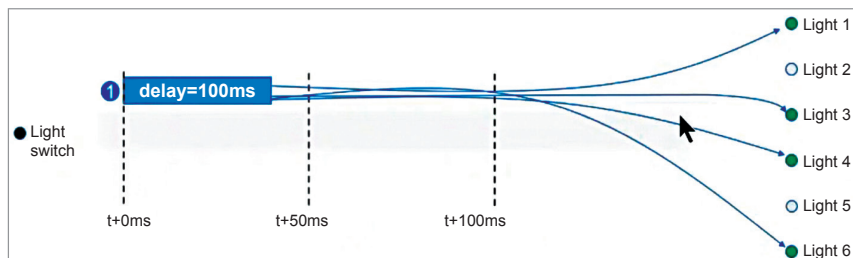


Figure 12: Message being transmitted three times at 50ms intervals

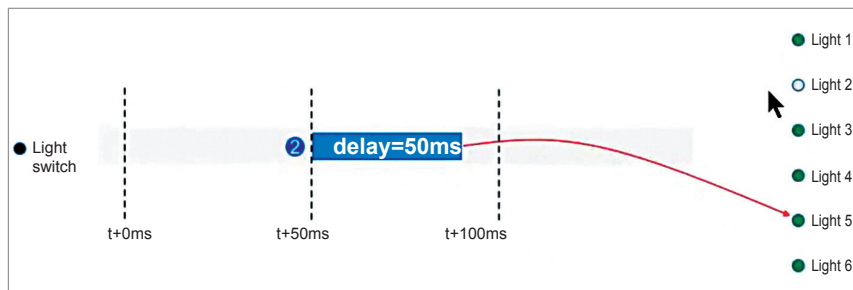


Figure 13: Second copy is sent 50ms later and is received by the fifth of the six lights

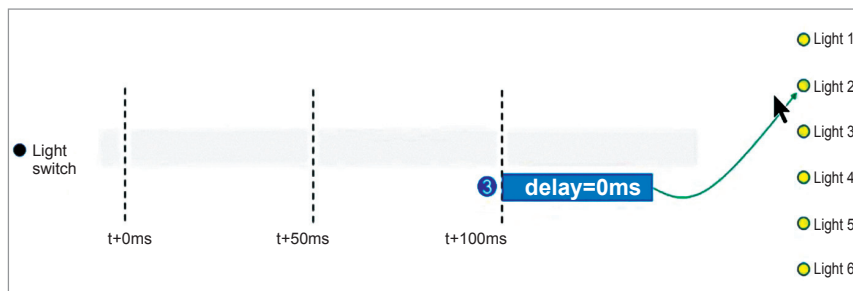


Figure 14: Final copy has no delay specified

Some key issues about Bluetooth mesh

Advertising bearer. The main bearer used in Bluetooth mesh is connectionless and, as such, there is no mechanism for coordinating with the scanning and advertising schedules of other nodes in the network.

Rx duty cycle. To maximise the probability of messages being received, the Rx duty cycle should be high. However, a Rx duty cycle of 100% is rarely achievable.

Acknowledgements. Acknowledged messages can be useful in one-to-one communication but do not scale in multicast messaging to groups of arbitrary sizes; so unacknowledged messages should normally be used.

Stochastic system. Includes inherent randomness and unpredictability.